ORION
Technical Proposal

Submitted for Participation in

ORION

UAV Design Challenge

By

TEAM Zephyr_Senior





Indian Institute of Space Science and Technology
Thiruvananthapuram, India

10-11 October 2026

Chapter 1

Team Composition

1.1 Team Information

Team Name: Zephyr_Senior
Institute: Indian Institute of Space Science and Technology (IIST)

1.2 Team Members

Aditya Chincholkar B.Tech Electronics and Communication Engineering Engineering, 2029 Team Leader	**Naman Gandhi** B.Tech Electronics and Communication Engineering, 2029 Team Member
Prathmesh Pachling B.Tech Aerospace Engineering, 2028 Team Member	**Rudra Pratap Singh** B.Tech Electronics and Communication Engineering, 2029 Team Member

Table 1.1: Team Zephyr_Senior members

Chapter 2

Problem Statement

2.1 Problem Statement

Restore temporary 4G or LoRa connectivity over a disaster area using a tethered or free-flying relay platform.

2.2 Objectives

- **Endurance:** Achieve a minimum flight time of 4hours with a 2kg communication payload.
- **Structural Integrity:** Design a robust frame optimized for maximum structural integrity, capable of withstanding the required flight maneuver limit load and wind speed of 10m/s.
- **Coverage:** Ensure connectivity over a 3km coverage radius.
- **Fail-safe Protocol:** Implement a fail-safe protocol to trigger immediate Return-to-Land (RTL) upon telemetry link disruption or critical voltage drops.

2.3 Mass Budget

Accurate mass tracking is critical to ensure the UAV stays within its regulatory and performance thresholds. The table below details the component mass breakdown.

Component	Quantity	Unit Mass (g)	Total Mass (kg)
Motors	6	240	1.44
Frame Tubes	12	111	1.332
Frame Rods	12	38.3	0.4596
Base Plate + Top Plate	1	212	0.212
Motor Clamps	6	30	0.18
Flight Computer	1	73	0.073
Wires	10.8	29.6	0.31968
ESC	6	109	0.654
Power Distribution Board	1	150	0.15
Buck Converter	1	100	0.1
GPS	1	12.2	0.0122
LoRa Module	1	8	0.008
Propeller	6	84	0.504
Battery	1	2500	2.5
MISC	1	400	0.4
Tether	30	23.33	0.6999
Payload	1	2000	2
		Total	**11.04438**

Table 2.1: UAV Mass Budget Breakdown

Accordingly, we have selected ”Vector Technics 6010 PRO” Motor which has a thrust of 2748g at 50% throttle, ensuring our Motors can support and hower the drone while taking into account wind speed of 10m/s.

2.4 Power Budget

Component	Voltage (V)	Hover Current (A)	Hover Power (W)
Motor (6x combined)	48.8	45.6	2225.28
Flight Controller	4.1	2.5	10.25
GPS & Telemetry	3.3	0.04	0.2
Total System	–	–	**2235.73 W**

Table 2.2: Electrical Power Budget

We are using a 400V - 7.5A, 13AWG wires to supply current from our ground generator to the DC-DC convertor present on our drone which will step down the voltage to 48V. This ensures we have enough power to supply the drone flight.

$$\text{Hover Endurance (Hours)} = \frac{\text{Battery Capacity (Wh)}}{\text{Hover Power (W)}} \times \text{Discharge Limit (80\%)}$$

Since, we are using a tethered-based system, Hover Endurance does not apply. Instead, a back-up battery present on the drone will be autonomously switched to in-case of abrupt power cutoff.

2.5 Conclusion

The proposed UAV design meets all fundamental criteria for the hackathon challenge. Through the integration of high-strength components and a proper power budget, our team is positioned to achieve high operational reliability and mission success.

Annexure B — Participant Undertaking

We, the members of the crew named below, confirm that we have read and understood the ORION Rulebook v1.0 and agree that we will:

[leftmargin=1.5em, itemsep=2pt, topsep=4pt]

- do our own original work and make honest declarations;
- follow the Ethical Code and the venue rules, and treat everyone with respect;
- accept the judges' decisions and the disciplinary process, including the organizers' right to disqualify for breaches;
- allow photography and recording for event documentation and, if our crew wins, the sharing of our portfolio with partner companies (14.4);
- accept that the organizers may amend the rules, announced through the official channel.

Annexure C — Pre-Built Module, Third-Party Material & AI/Tool Declaration

Submit before Kickoff (8.3) and update with the final submission. Types: pre-built module, third-party model, library or software, AI tool.

Crew name: _____ Track: _____

We declare that this list is complete and true. Team Leader's signature:

_____